

AI-Augmented Demand Forecasting Engine Design (v3.29)

Nature of this document: Cross-domain methodology paper spanning Time-Series Analysis / Machine Learning / ERP / AI, describing erpilot v3.29's demand forecasting engine. This release closes the upstream gap in the v3.25.10–v3.28 planning chain by replacing customer-manual MPS entry with automatic forecasts from historical sales + LLM augmentation.

■ Prerequisites: [MRP_ALGORITHM_DESIGN_EN.md](#) (v3.25.10) / [THROUGHPUT_ACCOUNTING_DESIGN_EN.md](#) (v3.28)

Abstract

erpilot's v3.25.10 → v3.28 planning chain assumed MPS is entered manually by the customer, undermining the objectivity of the entire IE/OR chain. This paper describes v3.29's demand forecasting engine, integrating Croston's (1972) classical intermittent demand algorithm, the Brown (1959) SES / Holt (1957) / Winters (1960) exponential smoothing family, and Hyndman & Koehler's (2006) MASE metric for automatic best-method selection. We also introduce Chatfield's (1989) regime change detection as an LLM-augmentation hook — when recent residuals exceed $\pm 2\sigma$, the system prompts customers to apply business judgment. Makridakis et al. (2018, 2020) M4/M5 competitions demonstrate that classical methods remain competitive with — sometimes beat — deep learning on most series, so this release excludes scipy/statsmodels/torch dependencies, implementing 5 methods in pure Python. 30 structural-invariant tests pass, including Hyndman-Koehler 2006's classic property $\text{MASE} < 1 \Leftrightarrow \text{beats naive baseline}$.

Keywords: Demand forecasting, Croston, Holt-Winters, MASE, intermittent demand, regime change, LLM augmentation

1. Introduction

1.1 The Fundamental Problem of Manual MPS

The 4 sprints v3.25.10 → v3.28 completed MRP-II, CLSP, and the TOC trilogy. But all plans start from an MPS that requires customers to fill in numbers like "produce 100 units of product A in W22." In practice:

- SMB owners lack data-analysis capability

- Sales history **exists**, but customers don't know how to use it
- Most parts have **intermittent demand** (zero for many weeks, then a spike) — intuitive entry is unreliable
- v3.27 explainability: "Why do we need so many M6 bolts next week?" always traces back to MPS — which the customer filled in, so erpilot can't ask 'are your numbers right?'

1.2 Why Not Deep Learning Directly

Makridakis (2018) M4 Competition results were striking: on a blind test of 100,000 time series:

Method category	Average sMAPE	Rank
Combine (weighted average of classical)	11.4%	1st
Exponential smoothing + ARIMA	11.9%	2nd
Deep learning (LSTM, MLP)	13.7%	bottom
Naive	14.3%	baseline

Makridakis (2020) M5 using Walmart sales data: classical and ML perform similarly, but **classical is more robust, more interpretable, and cheaper to train**.

For SMB (short history, mostly intermittent), classical **wins more often**. We therefore choose:

-  Naive / Seasonal Naive baseline
-  SES / Holt / Holt-Winters
-  Croston (intermittent specialist)
-  No Prophet / N-BEATS / TFT for now (deferred to future sprints)

1.3 Why Croston Is Irreplaceable

Croston (1972, *J Op Res Soc* 23) founded the methodology for "intermittent demand forecasting." In ERP scenarios:

- Industrial screws: 0-0-0-50-0-0-30-0 per week
- Machine spare parts: once every 3 months
- Seasonal gifts: only November orders annually

SES **always over-forecasts** such data (low-bias error assumption fails). Croston decomposes:

$$\text{forecast} = (\text{smoothed_demand_when_nonzero}) / (\text{smoothed_interval_between_demands})$$

Syntetos & Boylan (2005) proved the original Croston has positive bias, requiring a $(1 - \alpha/2)$ correction. We default to this corrected version.

2. Methodology

See [DEMAND_FORECASTING_DESIGN_ZH.md](#) §2 for full mathematical formulations of all 5 methods (Naive, SES, Holt, Holt-Winters, Croston) and accuracy metrics (MAPE, sMAPE, MASE per Hyndman-Koehler 2006).

Key design choices:

- **Auto-selection** by lowest MASE on holdout
- **Intermittent detection**: when `zero_fraction > 40%`, restrict candidates to {Naive, Croston} (per Syntetos-Boylan 2005)
- **Prediction intervals** via residual bootstrap, $PI_{t+k} = \hat{y}_{t+k} \pm z_{\alpha} \sigma \sqrt{k}$
- **Regime change flag**: k of recent N residuals beyond $\pm \theta \sigma$ (Chatfield 1989 §13)

3. Implementation

```
backend/app/services/demand_forecasting.py    (~600 lines, pure Python)
├─ ForecastMethod (Enum: NAIVE / SEASONAL_NAIVE / SES / HOLT / HOLT_WINTERS / CROSTON / AUTO)
├─ ForecastResult / BacktestResult / RegimeChangeAlert (dataclasses)
├─ 5 forecasting algorithms (each ~30-80 lines)
├─ 3 accuracy metrics (MAPE, sMAPE, MASE)
├─ backtest / auto_select_method
├─ prediction_intervals_residual
├─ detect_regime_change (LLM augmentation hook)
└─ forecast (top-level API)
```

4. Validation

4.1 30 Structural Invariant Tests (10 categories, all pass)

Category	Tests	Coverage
1. Naive baseline	4	Flat output / empty / seasonal rotation / fallback
2. SES (Brown 1959)	3	$\alpha=1$ tracking / $\alpha\approx 0$ smoothing / constant recovery
3. Holt (1957)	2	Linear extrapolation / horizon linearity
4. Holt-Winters (1960)	2	Perfect cycle recovery / short-history fallback
5. Croston (1972)	4	All zero / rate recovery / flat output / single event
6. Accuracy metrics	6	MAPE / sMAPE / MASE classic properties
7. Auto-selection	2	Intermittent restriction + lowest MASE wins
8. Prediction intervals	2	Widens with horizon / zero residual zero width
9. Regime change	2	Detect spike / no false positive on stable
10. API integration	3	Full ForecastResult / horizon=0 / explicit method

4.2 Results

30/30 tests pass. Sprint cumulative: 442/442 smoke tests pass.

5. Limitations and Future Work

Topic	Why not	Future direction
Modern ML (Prophet / N-BEATS / TFT)	M4 shows classical still competitive; heavy deps	v3.30+: optional deps, GPU env only
Time-series CV	Holdout only, not rolling	Hyndman 2021 §5.4 rolling CV
Hierarchical reconciliation	product/category/total inconsistency	Wickramasuriya, Athanasopoulos & Hyndman (2019) MinT
Full auto-tuning of (α , β , γ)	Only α grid search	Nelder-Mead via scipy
Multiplicative seasonality	Additive only	Holt-Winters multiplicative variant
External regressors	Univariate only	ARIMAX
ETS state-space framework	Not integrated	Hyndman, Koehler, Snyder & Grose (2002)
Actual LLM augmentation	Spec defined, API not wired	v3.30: prompt template + ConfirmCard flow

Edge Cases

1. History < 2 seasons: HW auto-falls-back to Holt
2. All-zero history: all methods return 0
3. Single non-zero event: Croston uses mean / expected interval
4. Negative demand (returns): not specially handled; recommend "net sales" input

6. Legal Notice / 法律聲明

Important: Legal Nature of Statistical Forecasts

This module produces statistical demand forecasts. All forecasts carry inherent uncertainty and are NOT:

1. **Guarantees of future demand:** point forecasts are algorithm-computed expected values based on historical data; actual demand may differ significantly. **Prediction intervals** are

probabilistic approximations — a 95% PI does **NOT** guarantee 95% capture rate (especially when data-generating process changes)

2. **Basis for procurement commitments:** customers **must NOT** use this module's forecasts as the **sole basis** for supplier commitments, employee overtime, or customer delivery guarantees. Major decisions must integrate market conditions, specific customer SOs, and business intelligence
3. **Evidence in supply-chain disputes:** this module's output **must NOT** be cited as "the forecast said X, so the supplier should have..." in disputes. Forecasts are fundamentally **estimates**, not contractual commitments
4. **Prediction of unforeseeable events:** no statistical method can predict:
 - Natural disasters (earthquakes, floods, typhoons)
 - Policy changes (tariffs, regulations, trade wars)
 - Customer business changes (bankruptcy, mergers, relocation)
 - Competitive landscape shifts (new entrants, technology breakthroughs)
 - Black Swan events (Taleb 2007)

1. Algorithm Limitations

- **Stationarity assumption:** all methods assume the data-generating process is stable within the forecast horizon. **Regime change** detection is a rough indicator, NOT a replacement for business judgment
- **MASE < 1 doesn't guarantee correct decisions:** MASE is "relative to naive baseline"; MASE 0.5 may still be insufficient in absolute accuracy for major decisions
- **Mis-calibrated PIs:** when residuals exhibit autocorrelation or heteroscedasticity, PIs are inaccurate
- **Auto-selection overfitting:** choosing methods by holdout MASE may be "lucky" on holdout but unstable in production

2. LLM Augmentation Warning (when enabled in v3.30+)

When `detect_regime_change` triggers LLM review:

- LLM may **hallucinate** business reasons (fabricate customer/competition scenarios)
- LLM's qualitative adjustments should be **advisory only**, never auto-overwriting forecasts
- All LLM suggestions must go through **ConfirmCard** (human confirmation)
- LLM-stated reasons (e.g., "Customer X cancelled") **do NOT constitute factual statements about Customer X** — avoid defamation risk

3. Croston / Intermittent Demand Special Notice

Croston (1972) assumes **non-zero demand events** are a random process. In practice:

- If demand events have **structural causes** (e.g., end-of-month inventory top-up), statistical methods may still fail
- Syntetos-Boylan (2005) bias correction is correct in mean but still biased in quantile estimation
- For **extremely sparse data** (e.g., once-per-year), no statistical method is reliable — use business intelligence

4. Responsibility for MPS Integration

This module can auto-generate MPS suggestions. But:

- **MPS final authority should rest with the production controller**, not be auto-generated
- Workflow must follow **ConfirmCard**: forecast → MPS suggestion → controller review → confirm → flow into MRP
- **Major decisions** (annual bulk material procurement, capacity investment) require additional review

5. Disclaimer Clause

To the maximum extent permitted by applicable law, erpilot assumes no liability for:

- Over-purchasing / line stoppage losses from acting on these forecasts
- Customer delivery defaults / penalties from forecast inaccuracy
- Erroneous business judgments influenced by LLM suggestions
- Planning failures from undetected regime changes
- Any consequences from using auto-generated MPS without controller review

6. Cumulative Applicability of Predecessors

This version completes the upstream gap of the v3.25.10 → v3.28 planning chain. **All disclaimers** in predecessor §6 documents **apply cumulatively** here.

Recommended Practice

- Treat forecasts as "**starting suggestions**" not "final MPS"
- Compare actual vs forecast monthly; update method selection and parameters
- High-value / high-risk parts (Class A) must have human review
- Demand outside PI → proactively contact customer to verify cause
- Retain forecast-accuracy history as reference for negotiation but **not as legal evidence**

7. References

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8. Changelog

Version	Date	Change
v3.25.10	2026-05-20	MRP-II (based on manual MPS)
v3.26 - v3.28	2026-05-20	CLSP / TOC trilogy (still on manual MPS)
v3.29	2026-05-20	This release: automatic demand forecasting engine, closes MPS upstream gap
Future v3.30+	TBD	LLM augmentation integration / Prophet evaluation / Hierarchical
Future v3.31+	TBD	Rolling CV / ETS state-space / Multiplicative seasonality

Last updated: 2026-05-20 (v3.29) **Authors:** erpilot engineering team (with TS analysis / ML / ERP / AI cross-domain academic methodology) **Version:** 1.0 **Chinese version:** [DEMAND_FORECASTING_DESIGN_ZH.md](#)